

Overcoming Autoregressive Myopia: Parallels from Computer Chess for Multi-Agent LLM Architectures

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Abstract

Large language models are fluent, confident, and often shallow. This shallowness is not a failure of capability but a structural consequence of monolithic generation. During autoregressive inference, attention dilution, the inability to backtrack, and single-path sampling jointly act as an implicit discount factor that makes distant constraint violations functionally invisible—even though the training loss weights all tokens equally. This mirrors the pathology of early chess engines, which grabbed pawns with shallow tactics while neglecting the slow, coordinated development that wins games. The analogy is refined to show that modern chess engines achieve depth through a deterministic move generator, a learned king-safety evaluator, and a search algorithm—and that language models need analogous components to overcome their present bias. Preliminary prompting experiments, including topological metrics of reasoning depth, suggest that the effective objective can be shifted at inference time to reward patient, coordinated reasoning. A research programme is then proposed that distributes reasoning across six specialised roles—a generator, an aesthetic critic for structural harmony, an analytic critic for constraint pressure, a deterministic verifier, an adversarial attacker, and a coordinator, with the neuro-symbolic translation step handled by a recursive instantiation of the same six-role architecture—whose coordinated interaction makes constraint satisfaction a constitutive requirement of generation at the system level. This architecture structurally parallels Global Workspace Theory, a leading cognitive model of consciousness, by implementing a functional analogue of a global workspace: the system monitors its own reasoning trajectory and intervenes when it detects internal contradictions. A parallel pattern appears in biology, where complex tasks are accomplished by networks of specialists, though the assembly mechanism differs. This architectural shift reframes the path toward more general deliberation capabilities within domains where constraints can be formally specified. The architecture applies to tasks where constraints can be formalised and where the constraint state can be structured to capture relevant partial decisions; fully general reasoning with arbitrary interlocking semantic constraints is beyond the current scope and flagged as future work. The chess engines found their path from pawn-grabbing to strategic dominance. The language models can too—but the architecture must change, not just the prompt.

Keywords: large language models, reasoning depth, architectural discounting, chess engine analogy, constraint satisfaction, multi-agent reasoning, Global Workspace Theory, inference-time compute, beam search, neuro-symbolic architecture

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1. The symptom

Ask a large language model to produce a framework that must satisfy twelve interlocking, non-negotiable constraints—apodictic certainties, reflexive closure, the hard problem of consciousness—and it will deliver an answer. The answer will be structured, confident, and fluent. It will also be shallow. The hardest constraints will be handled with a gloss, or quietly dropped. The reasoning will be locally coherent but globally brittle. Question it, and it will reassure you that all is well, because the training distribution contains many examples of people concluding their work is sound and few examples of people genuinely finding and repairing deep contradictions.

This failure mode is pervasive. It has been called hallucination [1], sycophancy [2], and shortcut learning [3]. Those labels describe symptoms. They do not explain the mechanism.

2. The diagnosis: architectural discounting and the horizon effect

To understand why language models produce shallow reasoning, we must look past the training objective and examine the generation process itself.

During pre-training, the model optimises a cross-entropy loss over the entire sequence. Mathematically, there is no temporal discount factor; a mistake at token 5,000 is weighted identically to a mistake at token 10. Superficially, this suggests the model has no incentive to ignore long-range dependencies. However, the *loss function* is not the same as the *generation process*. At inference time, the model generates tokens autoregressively, conditioned on all previous tokens, but with no ability to revisit earlier decisions. The attention mechanism—the primary conduit for long-range information—dilutes with distance; information in the middle of long contexts can be effectively lost [4]. The model cannot backtrack to repair a contradiction introduced three thousand tokens ago. Single-path sampling means any early choice that later proves inconsistent cannot be undone.

The net effect is an **architectural discount factor**—not a tuned parameter, but an emergent decay in the relevance of distant constraints. The model functionally discounts future coherence because it cannot reliably attend to it, cannot revise past decisions in light of it, and cannot explore alternative paths that might satisfy it.

2.1. Information-theoretic capacity ceiling

The generation process can be formalised as a Markov chain on the sequence prefix: if $S_t = (X_1, \dots, X_t)$ denotes the prefix up to token t , then the next token depends only on S_t , so $C \rightarrow S_t \rightarrow X_{t+1}$ is a valid Markov chain, where C represents the original constraints. By the Data Processing Inequality (Cover & Thomas, 2006 [35]), we have $I(C; X_{t+1}) \leq I(C; S_t)$. This provides an absolute information-theoretic ceiling: a single generated token cannot contain more constraint information than the full prefix already holds. The DPI does not guarantee that this ceiling is reached; empirical attention dilution imposes an additional mechanical bottleneck, pushing the effectively accessible information far below the theoretical maximum. Distant constraints thus become functionally invisible, not because the prefix lacks the information, but because the attention mechanism cannot route it to the current prediction. The DPI guarantees that this dilution, when it occurs, cannot be compensated by the next token containing *more* information than the prefix—but the prefix itself can be augmented by explicit re-signalling, which is precisely what the architecture’s critics and verifier are designed to enforce.

2.2. The Horizon Effect

This architectural discounting produces a behavioural pathology analogous to the **Horizon Effect** in early chess engines. In game AI, an engine that cannot calculate a defence against an impending attack within its search depth will execute a series of harmless delaying moves to push the disaster just past the horizon, where it becomes invisible. This linguistic Horizon Effect is a direct analogue to the search pathology formally defined in classical heuristic search [5], where an agent executes delaying actions to push unavoidable negative consequences beyond its fixed search depth. LLMs exhibit the same behaviour: when approaching a hard constraint they cannot satisfy, they generate fluent, topically relevant filler paragraphs to push the violation past their immediate cognitive horizon. The model delays the inevitable contradiction until it hits the context limit or triggers the end-of-sequence token, leaving the violated constraint—the mated king—just over the horizon, never explicitly addressed (Figure A.1).

The analogy is functional, not mechanistic: the LLM does not compute that a contradiction is imminent and then strategically delay it. Rather, its attention horizon prevents it from ever representing the contradiction clearly; the filler is a consequence of blindness, not of foresight. The structural result—a locally coherent output that globally violates a constraint—is the same.

2.3. Formal grounding

Pearl (1984) [5] proved that in any finite-horizon, evaluation-guided search, there exist positions where the optimal shallow move differs qualitatively from the true optimal move, causing the agent to execute delaying actions that push an unavoidable loss past the search horizon. This result maps directly to autoregressive LLM generation when modelled as a one-player search over reasoning states: the model, operating under a fixed attention horizon and a myopic evaluation signal (fluency, local coherence), will—for some constraint sets—produce fluent, locally coherent filler that postpones a contradiction until it falls outside the effective horizon, rather than resolving it. The Horizon Effect is therefore not a quirk of early chess engines; it is a structural property of any fixed-horizon search with an imperfect evaluation function.

Reinforcement learning from human feedback (RLHF) compounds the problem. Human raters generally evaluate the final output or short snippets. They prefer helpful, concise, confident answers and penalise hesitation, uncertainty, and prolonged exploration [6]. When the RLHF reward model provides a terminal signal that ignores intermediate reasoning, the generation process becomes even more myopic: the immediate reward of emitting a fluent, conclusive token overrides the deferred, architectural need for global coherence. This myopia is exacerbated by reward model overoptimization. As demonstrated by Gao et al. [7], when models are optimized too heavily against a proxy reward model, they inevitably exploit its blind spots—a manifestation of Goodhart’s Law.

What are the “pawns” in this RLHF framework? Pawns are the superficial heuristics that human raters and reward models easily recognise and reward: markdown formatting, polite hedging, confident tone, and the appearance of comprehensiveness. The model learns to aggressively grab these pawns—optimising for surface-level helpfulness—because they yield immediate reward. It will gladly sacrifice the king (deep logical consistency and constraint satisfaction) to secure a beautifully formatted, highly confident, but hollow response.

The Weak-Link Constraint. In short, the model exhibits a structural bias towards the here and now—analogue to human present bias, long studied in reinforcement learning [5]. This bias is a direct consequence of the monolithic architecture. A single model must simultaneously satisfy fluency, constraint adherence, consistency, and completeness within one forward pass. If any cognitive function is weak, that function caps the performance of the entire system. This structural limitation follows directly from the probability axioms: the event that the chunk is valid implies satisfaction of each requirement i , so its probability cannot exceed $\min_i q_i$, where q_i is the probability that requirement i is satisfied. We refer to this as

the weak-link constraint: monolithic generation is bounded above by its most neglected cognitive function. The monolithic LLM is the architectural equivalent of a chess player who develops only one piece and then finds that piece can only do so much when its allies cannot join the attack.

Empirical benchmarks confirm this architectural limitation: despite their vast parametric knowledge, current LLMs consistently fail at tasks requiring explicit, multi-step planning and reasoning about state changes [8], confirming that the deficit lies in the generation mechanics rather than the knowledge base.

This dual myopia—architectural discounting during generation, compounded by short-term reward shaping during RLHF that explicitly rewards surface polish over substance—explains why LLMs produce shallow reasoning. It is not a failure of capability. It is a failure of the generation architecture and the evaluation function that guides it.

3. The chess engine analogy, refined

The history of computer chess illuminates both the problem and the shape of a possible fix.

Early engines: shallow search and rigid king safety. The earliest chess engines, proposed by Shannon in 1950 [9], evaluated positions using handcrafted heuristics: count material, reward castling, penalise moving the pawns in front of the king. These rules were not foolish, but they were rigid. An engine might reject a winning pawn storm because the king’s pawn shield looked intact by its rules, or cling to a castled position while a dynamic attack gathered. Moreover, early engines searched only a few ply deep. A threat twelve moves away was functionally invisible—not because the evaluation function assigned it zero weight, but because the search architecture could not reach it. The engine grabbed pawns because those were the gains visible within its horizon, and it suffered from the Horizon Effect—pushing inevitable losses just past the search depth. The lone knight, hunting a rabbit, had no sense that it was exposing the king.

Modern engines: deep search and learned, fluid king safety. Modern engines like Stockfish and AlphaZero evaluate king safety with a neural network trained on millions of game outcomes [10, 11]. They search millions of positions, evaluating each with a learned value function that predicts the probability of eventual failure. They feel the pressure building even when no simple rule is violated. They recognise an impending attack and weight that danger so heavily that it overrides material gain. Their search depth allows them to see threats far beyond the immediate horizon. A move that wins the queen but allows a forced checkmate thirty ply later is scored as losing.

To achieve this, modern chess engines rely on three distinct components. First, a **deterministic move generator** enforces the rules of the game: only legal moves are ever considered. Legality is not estimated; it is a hard boundary. Second, a **learned evaluation function** scores positions by combining multiple signals. Two signals are particularly instructive. *Piece mobility* rewards active, well-coordinated pieces even when no immediate material gain is visible. A knight moving from a1 to c3 gains scope; it controls more squares and can participate in future attacks. This is a forward-looking proxy: better-developed pieces increase the probability of a decisive win, even if the exact winning line is beyond the search horizon. *King safety* evaluates the opponent’s attack potential around the king—how many pieces are pointing at it, how many escape squares remain, how exposed it is. This is a backward-looking proxy from the objective: the engine asks what must be true for the king to fall, and scores positions by how close they are to that condition. Neither signal alone is sufficient. Together they illuminate a corridor (Figure A.2): the forward pass from piece mobility meets the backward pass from king safety, and the search navigates between them. This bidirectional evaluation structure dramatically reduces the effective branching factor compared to forward-only evaluation, guiding the forward search with both a pull from the objective and a push from structural harmony.

Third, a **search algorithm** (minimax with alpha-beta pruning, or Monte Carlo tree search) explores alternative futures, using the evaluation function to score leaves and the move generator to define the branches. The search is the coordinator of the other components; it is where forward and backward signals meet.

The aesthetic dimension of coordination. Piece mobility and coordination are not only measurable by analytic definitions—squares controlled, files opened. They are also perceptible through a different cognitive channel: their harmonic beauty. A well-developed chess position looks beautiful to an artistically inclined observer. The pieces are not randomly placed; they relate to one another in a way that is immediately pleasing. This aesthetic perception is not a superficial ornament. It is a legitimate mode of knowing, parallel to analytic evaluation, that grasps the whole without decomposing it. The artist sees the same coordination that the analyst quantifies; each perceives truth through a different faculty.

This mapping of cognitive faculties to chess evaluation is grounded in a long tradition of research on expert perception. De Groot [12] and Chase & Simon [13] demonstrated that grandmasters perceive chess positions in larger chunks—meaningful configurations rather than individual pieces—and that this chunked perception is what distinguishes experts from novices. Intuition, in this tradition, is not mysticism; it is the rapid, pattern-based recognition that comes from thousands of hours of exposure. The aesthetic appreciation of a well-coordinated position is a natural extension of this chunking capacity: the grandmaster not only recognises the configuration but perceives its quality holistically, as a single impression. Logic, by contrast, operates step-by-step, each inference a single move. Both modes are valid. Each covers the other’s blind spot. Mature human reasoning moves fluidly between them. Expert chunking is, however, strictly domain-specific: grandmasters chunk chess positions, not arbitrary patterns. An aesthetic critic that relies on chunked pattern recognition would therefore require domain-specific training for each reasoning domain. This is feasible for the scoped programme of verifiable-domain reasoning but constrains the critic’s generality.

What this means for language models. In an LLM reasoning task, the equivalent of the king is the set of all explicit and implicit constraints a correct solution must satisfy. A mated king is a violated constraint—an unaddressed requirement, a buried contradiction. King safety is a dynamic estimate of how close the current generation path is to violating one of those constraints. Piece mobility is a measure of conceptual coordination: are the primitives well-defined, are the derivations connected, are the constraints being addressed or merely gestured at?

Current LLMs, under their default generation architecture, represent early chess engines. Their single-path autoregressive generation acts as a shallow search. Their evaluation function—shaped by RLHF—rewards local fluency, the cognitive equivalent of material. They may have some rigid “safety” heuristics: avoid obvious contradictions, sound appropriately cautious. But they lack a fluid, weighted sense of impending failure, and their generation horizon is too short to see the contradiction coming.

4. Preliminary evidence: shifting the effective objective at inference time

If part of the problem is the evaluation function and the generation architecture, then changing the effective objective at inference time—through prompting—should alter behaviour. The prompt acts as a context that reshapes the effective objective, mimicking a deeper, more constraint-sensitive evaluation by conditioning the model on tokens that signal the value of long-term coordination.

I tested this informally with a demanding systems-engineering task: designing a complex software architecture subject to twelve interlocking, non-negotiable constraints (e.g., strict dependency acyclicity, specific memory-allocation limits, and formal API contract adherence). Here, “constraint satisfaction” means that the proposed architecture explicitly addresses each requirement and passes deterministic static-analysis checks without internal contradictions.

The baseline behaviour, with no special prompting, was typical: a polished, confident answer under thirty seconds that satisfied none of the hard constraints.

I then tested a series of meta-prompts designed to impose a different implicit objective—one that penalises premature closure, rewards weak-point prioritisation, and requires the model to certify an exhaustion criterion before concluding.

- An **exploration-first protocol**, simulating adversarial coordination among multiple critical personas and requiring explicit certification that every weak point had been addressed, produced **445 seconds** of sustained, self-critical reasoning. The output was qualitatively different: extended internal debate, explicit tension logging, repeated backtracking, and genuine engagement with the hardest constraints.
- A **puzzle-structured protocol**, embedding the task as adversarial empirical puzzles, produced only **107–174 seconds**. Abstract constraints did not generate the same internal friction.
- A **“Stockfish-level” protocol** produced intermediate results around **257–338 seconds**, varying with specific wording.

The key finding: a carefully designed prompt that explicitly penalises pawn-grabbing and demands coordination can more than double sustained reasoning time, with corresponding qualitative shifts. This supports the diagnosis that a significant part of the shallowness is evaluation-driven.

Beyond thinking time: topological metrics of reasoning depth. While thinking time serves as an initial proxy, it is imperfect, as models can generate verbose filler without genuine progress. Therefore, I also examined the structural topology of the outputs, specifically tracking **Backtracking Frequency** (explicit revision of prior claims, marked by phrases like “Wait,” “Actually,” or “Let me reconsider”) and **Constraint Interweaving** (whether the constraints were addressed in isolated, sequential blocks or synthesised dynamically throughout the response). The baseline condition exhibited zero backtracking and strictly sequential, isolated constraint handling—each constraint addressed once in a single paragraph and never revisited. In contrast, the exploration-first protocol triggered frequent backtracking events and dynamic interweaving, with constraints revisited multiple times as new insights emerged. This demonstrates that the prompt successfully shifted the model from a greedy, single-path search to a deliberative, self-correcting trajectory (Figure A.4).

These are informal, single-model, single-task observations. They are not controlled experiments. I present them as existence demonstrations, not proof.

5. The principle: coordinate the army, reward the development

These preliminary results suggest that the effective objective can be reshaped by prompting. The next question is: what should that objective optimise? The chess analogy yields a precise answer.

In chess, involving more pieces in an attack increases the probability of a decisive win—winning material, and ultimately the king. However, coordination takes moves. Improving a poorly placed piece requires patience and yields no immediate material advantage. A shallow search cannot see the eventual payoff; it sees only the cost of a move on a piece that is not yet contributing.

The correlation is therefore: more coordinated pieces → greater potential gains, including the ultimate gain of the king. But because coordination demands depth, a myopic evaluation function paired with shallow search systematically undervalues it. The engine grabs the pawn with the already-active knight, because the deeper line lies beyond its horizon.

For LLMs, the same logic applies. The “weak links” are the constraints that have not yet been adequately addressed—the reasoning equivalent of undeveloped pieces. Addressing them yields no immediate fluency reward. But only by addressing them can the model bring its full conceptual army to bear on the central problem, satisfying all constraints simultaneously—the cognitive equivalent of delivering checkmate.

The practical implication is that we must incentivise the model to reward incremental piece improvement *even when no immediate gain is visible*. In chess engine terms, this means strengthening two distinct but complementary evaluation signals:

1. **King-safety evaluation (constraint-pressure evaluator):** A function that heavily weights coordinated threats against the objective. In LLM reasoning, this scores the degree to which the current trajectory is accumulating unresolved tensions, constraints left unaddressed, or contradictions looming. It provides the backward-looking gradient from the necessary conditions of a valid solution.
2. **Piece-mobility evaluation (conceptual-coordination evaluator):** A function that directly rewards improving the scope and connectivity of concepts, independent of immediate payoff. In LLM reasoning, this scores the cognitive work of clarifying vague definitions, forging missing connections, and revisiting neglected constraints. It provides the forward-looking gradient from the current state toward a more coordinated framework.

The rhythm of reasoning. In the opening phase of a chess game, most pieces are undeveloped and the king is exposed. A player who chases pawns in such a position is punished. The correct default is to develop the army, coordinate the pieces, and attack only once the position is mature—unless a genuinely low-risk, high-reward tactical opportunity presents itself. An artist or architect works the same way: the broad framework is established before the details are filled in.

The bidirectional scoring naturally reproduces this rhythm. When many constraints remain unaddressed, the Analytic Critic’s pressure V_{bwd} dominates the score, penalising neglect and pushing the search toward constraint-addressing chunks. A candidate that is fluent but ignores a weak link must clear a higher bar: the forward-looking harmony gain must outweigh the backward-looking pressure penalty. As constraints are progressively satisfied, V_{bwd} recedes, and the Aesthetic Critic’s harmony signal gains influence, allowing the search to exploit tactical completions and elegant resolutions. This shift reflects an underlying gradient, to a first approximation: the marginal benefit of addressing a neglected constraint decreases as the solution matures, while the probability of a low-risk, high-reward tactical completion increases as the reasoning army coordinates. The architecture therefore defaults to coordination in the early stages, shifts gradually toward exploitation as the solution matures, and permits tactical deviations at any point provided the critics jointly signal that the risk is low and the reward high.

Neither pure conservatism nor pure opportunism is robust to uncertainty: the Aesthetic Critic may miss a tactical threat, and the Analytic Critic may overstate a minor constraint. By always combining both signals, the Coordinator maintains a hybrid midground—it leans conservative when pressure is high, opportunistic when pressure is low, but never switches entirely to one mode. No external schedule or hyperparameter controls this rhythm. It emerges from the state-dependent interplay of the two evaluation signals. When the position is undeveloped, the pressure is high; when the army is coordinated, the pressure subsides. The search inherits the strategic cadence of the chess board itself.

Unstructured exploration as an architectural option. In chess, some players introduce a near-random element into their search—shuffling through unlikely continuations or allowing the mind to wander—to speed up the discovery of refutations or creative ideas that structured scanning alone might miss. The proposed architecture borrows this concept. The Generator’s temperature-modulated sampling and the Coordinator’s optional Monte Carlo rollouts provide a controllable source of unstructured exploration. The

cybernetic feedback loop activates this mode when pressure is low and the position is stable, ensuring that randomness serves discovery rather than introducing unnecessary noise. The architecture does not require this mode; it provides it as an augmentation that can be dialled up or down depending on the compute budget and the reliability of the critics.

Figure A.2 visualises how these two evaluation signals guide search.

6. A research programme: the society of specialised LLMs

The meta-prompts are a patch. They redirect the generation process at inference time, but the underlying architecture—the single-path autoregressive generation that functionally discounts distant constraints—remains unchanged. The deeper question is whether we can build systems whose generation process intrinsically values constraint satisfaction, just as a chess engine intrinsically respects the rules of the game.

A chess engine does not consult an external oracle to check whether a move is legal. Legality is constitutive of move generation: only legal moves are ever considered. The equivalent for language model reasoning would be to make constraint satisfaction a constitutive requirement of the generation process. The model should not be able to output a proposed solution without first verifying—through a mechanism that does not depend on the same fallible neural weights that produced the solution—that every constraint is satisfied. In an LLM, where legality is a complex semantic property, the Generator cannot be constrained internally. Instead, the architecture enforces constraint satisfaction **constitutively at the system level**: no output can be emitted without passing the Deterministic Verifier. The Generator may propose invalid chunks; the system discards them. This is analogous to a chess engine that generates pseudo-legal moves—some of which may leave the king in check—and then filters them during search.

This proposal structurally revives Marvin Minsky’s foundational “Society of Mind” framework [14], which posited that intelligence emerges not from a single, monolithic algorithm, but from the coordinated interaction of numerous specialised, simpler agents. While Minsky’s original agents were symbolic and hand-crafted, we propose achieving this society through the orchestration of specialised neural networks.

The evolutionary history of chess engines suggests a path. They did not achieve superhuman play by scaling up a single evaluation heuristic. They achieved it by integrating specialised components—move generator, evaluation function, search algorithm—each doing what it alone does well, their coordination producing something none could achieve individually. A parallel pattern appears in biology, where complex tasks are accomplished by networks of specialists covering each other’s blind spots [15], though the mechanism of assembly—natural selection with a clear fitness function—differs fundamentally from the engineered coordination proposed here.

The exploration-first protocol (Section 4) already demonstrated that a simulated multi-agent loop can improve depth. The architecture proposed here makes those roles explicit, external, persistent, and trainable.

Applied to language models, this principle suggests an architecture in which reasoning is distributed across **multiple specialised LLMs**, each fine-tuned or prompted for a distinct cognitive role, communicating through structured interfaces. Recall from Section 3 that modern chess engines rely on three core components: a deterministic move generator, a learned evaluation function, and a search algorithm. Our architecture mirrors this decomposition with six specialised roles, with the neuro-symbolic translation step handled by a recursive instantiation of the same six-role motif (Section 6.3):

- **Generator (the forward pass).** An LLM optimised for fluent, creative production. It generates candidate solutions in large chunks—paragraphs, argument sections, whole frameworks—mimicking the intuitive, pattern-based forward search of a chess engine. This operationalises the chunked, intuitive

mode of cognition described in Section 3: instead of reasoning token by token, it generates whole patterns at once, leaving step-by-step verification to other components.

- **Aesthetic critic (the piece-mobility evaluator).** An LLM trained to score the structural coordination of a partial solution without decomposing it item by item—perceiving conceptual harmony in a single gestalt, much as an artist appreciates the beauty of a well-developed chess position. This critic perceives the ugliness of undeveloped, disconnected reasoning—the conceptual equivalent of a knight stranded on a1 or a pawn structure riddled with weaknesses. It rewards structural harmony and piece coordination, not because beauty is an end in itself, but because well-coordinated positions historically lead to victory.

The Aesthetic Critic estimates the **probability of eventual success**—the state-value function $V(s)$ of the underlying Markov Decision Process. Let the terminal reward $R(s_T)$ be 1 if the Deterministic Verifier certifies a complete constraint-satisfying solution, and 0 otherwise. Then

$$V(s) = \mathbb{E}[R(s_T) \mid s_0 = s]$$

is the expected probability of success from state s . The Aesthetic Critic is trained to predict $V(s)$ from a compact vector of **graph invariants** extracted from the conceptual graph of the reasoning trace: algebraic connectivity, orphan-node count, average clustering coefficient, and a graph-neural-network embedding. Because these invariants are fixed-size and do not grow with trace length, the critic evaluates long-term structural promise globally, without attention dilution.

Lemma 1 (Conditional Sufficiency of Graph Invariants). *Let the conceptual graph extraction be faithful—**isomorphic graphs receive the same Verifier verdict**. If the graph invariants $I(G)$ are injective on the set of attainable graphs, then there exists a function $\hat{f}$ such that $\text{Verdict}(T) = \hat{f}(I(G_T))$ for every trace T . The invariants are a sufficient statistic for predicting the Verifier’s eventual outcome.*

This lemma provides the formal basis for the critic’s architecture. Finding a computable, complete invariant for general graphs is at least as hard as the Graph Isomorphism problem; practical implementations must use lossy, approximate invariants whose injectivity is not guaranteed. The Aesthetic Critic therefore learns a mapping that is only approximate, and its accuracy is an empirical question. Its output, $V_{\text{fwd}}(s)$, is a reward: higher values indicate greater likelihood of success. The critic is trained on rollout traces: for each partial state, the object-level society attempts completion, and the Verifier’s final verdict provides the binary training label.

Graph connectivity is a proxy for structural harmony, not a direct measure of logical validity. In isolation, it would be vulnerable to the same Goodhart-style gaming that afflicts RLHF reward models. In the proposed architecture, however, the Aesthetic Critic’s score is only used to rank candidate partial solutions that have already passed the Deterministic Verifier; invalid solutions are pruned regardless of their connectivity score. The critic’s proxy is therefore constrained by hard verification, reducing the risk of proxy overoptimization. In some domains, excessive connectivity may indicate tight coupling or circular reasoning rather than genuine coherence; the critic’s training must therefore adapt the connectivity metric to the domain’s structural norms. The extent to which connectivity correlates with genuine reasoning quality across domains remains an empirical question.

The conceptual graph is extracted from the reasoning trace using dependency and discourse parsers for formal and semi-formal reasoning domains. Where such parsers are unavailable, an LLM-based extraction step is used, with the caveat that this reintroduces attention dilution for very long traces;

fully general graph extraction remains an open problem. At inference time, the critic consumes the compact invariant vector computed by the Coordinator, not the raw reasoning trace. In the terms of Section 5, this is the primary piece-mobility evaluator.

- **Analytic critic (the king-safety evaluator).** A neural network fine-tuned to provide a scalar score of intermediate reasoning states for proximity to constraint violation. Like a chess engine’s evaluation function, it operates continuously at every node of the search tree, detecting accumulated pressure: unresolved tensions, constraints left unaddressed, contradictions looming. It provides a continuous gradient at intermediate steps, even when the violation itself is many tokens away—allowing the Coordinator to prune doomed branches early, long before they reach the final Verifier.

Where the Aesthetic Critic sees structural ugliness—the undeveloped knight, the fragmented argument—the Analytic Critic calculates specific threats. It is the king-safety evaluator, tracking how close each constraint is to violation and how much pressure is accumulating around the king. The critic receives the Coordinator’s **Constraint State Register** (CSR) together with the candidate chunk, ensuring global constraint visibility without attention dilution (see Constraint State Management below).

Unlike the Aesthetic Critic, the Analytic Critic does not estimate a success probability. It is trained to predict the **expected constraint violation cost**—a scalar that increases with unresolved tension, accumulated pressure, and budget depletion—from the CSR. Its output, $V_{\text{bwd}}(s)$, is a *penalty*: higher values indicate greater risk. Theorem 3 guarantees that the CSR contains the information needed to estimate this cost.

Because the two critics estimate complementary quantities—forward-looking reward and backward-looking penalty—their combination in the scoring function $\alpha V_{\text{fwd}} - \beta V_{\text{bwd}}$ is a principled trade-off: the Coordinator balances the opportunity of structural harmony against the threat of constraint pressure. This subtraction is mathematically coherent and directly mirrors the chess engine’s balance of piece mobility and king safety.

In the terms of Section 5, this is the constraint-pressure evaluator. It complements the Aesthetic Critic: the artist sees the harmony of the developing framework; the tactician watches for threats to the solution’s integrity.

Recent work on process reward models underscores the need for this split. Neural PRMs exhibit prompt sensitivity, step-label interference, and overthinking, with accuracy gaps of 10–50% on long-horizon reflective reasoning traces [20]. Verifiable Process Reward Models that anchor step-level checks in deterministic rules substantially outperform purely neural PRMs [37]. The Analytic Critic’s reliance on the Constraint State Register and the Deterministic Verifier follows this emerging consensus: the critic learns a gradient, but the Verifier provides the ground truth.

- **Deterministic verifier (the environment physics).** An external symbolic system—a Python interpreter, a theorem prover, a formal logic checker—that enforces hard constraints. In a chess engine, the rules of the game are baked into the move generator. In an LLM, the “move generator” is merely the tokenizer, which only enforces syntactic validity. To enforce semantic and logical validity, we must introduce an external environment that acts as the physics of the reasoning world. It receives candidate outputs, translates them into executable checks, and returns a binary verdict. **The verifier checks all hard constraints on the current partial state; it does not require the solution to be complete.** It is the architectural boundary that makes constraint satisfaction constitutive at the system level, forcing the Coordinator to backtrack and reject any branch where the “physics” of the logic are violated.

This design directly mirrors Verifiable Process Reward Models [37], which replace neural step-verification with symbolic checks and achieve up to 20% higher F1 scores on structured reasoning tasks. By making the Verifier an external, non-neural oracle, the architecture avoids the self-evaluation pathologies that limit purely neural reward models.

- **Adversarial attacker (the discrete prober).** Unlike the analytic critic which passively scores states, the attacker is an LLM tasked with actively generating explicit stress-tests: formulating edge cases, proposing counter-arguments, and highlighting specific rhetorical camouflage. It forces the Generator to actively defend its logic against discrete, targeted challenges, surfacing violations that a passive scalar score might miss. The attacker uses the CSR to identify weak points and generates counterexamples targeting specific constraints. Its function is to detect and surface violations that the generator might otherwise push past its attention horizon, directly countering the Horizon Effect diagnosed in Section 2.
- **Coordinator (the search algorithm).** The coordinator implements a constrained beam search with iterative deepening. At each step, it maintains a fixed-width beam of candidate partial solutions. Each candidate is scored by a weighted combination of the analytic critic’s constraint-pressure estimate and the aesthetic critic’s coordination score. Candidates that fail the deterministic verifier are immediately pruned. Candidates flagged by the adversarial attacker are either repaired—if the generator can resolve the flagged issue within a bounded retry budget—or discarded. The search depth is controlled by a computational budget parameter, and the coordinator implements iterative deepening: it first explores all branches to depth 1, then depth 2, and so on, ensuring that the most promising lines receive the deepest exploration. This is directly analogous to heuristic-guided search with constraint-based pruning in chess engines: the evaluation functions provide the node ordering, and the search algorithm provides the traversal strategy, while hard constraints (the Environment Physics) prune invalid branches before they are expanded. The coordinator translates between the chunked representations of the generator and the intermediate checks of the analytic critic. It implements the bidirectional evaluation described in Section 3: the generator performs the forward pass, guided by the aesthetic critic’s holistic evaluation of promise; the analytic critic performs the backward pass from the necessary conditions of constraint satisfaction; the coordinator navigates the corridor where these two signals meet.

Formally, the Coordinator treats the generation process as a Markov Decision Process (MDP). At each intermediate state s_t (a partial solution), the Coordinator must select the next reasoning chunk. It evaluates candidate states using a bidirectional objective function:

$$\text{Score}(s_t) = \alpha \cdot V_{\text{fwd}}(s_t) - \beta \cdot V_{\text{bwd}}(s_t) - \lambda \cdot C(s_t) \quad (1)$$

Where $V_{\text{fwd}}(s_t)$ is the Aesthetic Critic’s estimate of eventual success probability (the forward-looking piece mobility reward), $V_{\text{bwd}}(s_t)$ is the Analytic Critic’s estimate of constraint violation cost (the backward-looking king-safety penalty), and $C(s_t)$ is the computational cost of the branch. The search is strictly constrained by the Environment Physics: any candidate state where the Deterministic Verifier returns $V_{\text{det}}(s_t) = 0$ (a hard constraint violation) is immediately pruned, regardless of its $\text{Score}(s_t)$. This formulation transforms the abstract concept of “deliberation” into a computable, optimizable search trajectory.

This division of labour directly addresses the weak-link constraint: under-investing in any cognitive function leaves that function as the weakest link, just as neglecting a chess piece leaves it unable to support the attack. A single, undifferentiated model cannot bring all functions to bear simultaneously; the specialised society can.

Together, these six roles form a complete reasoning loop. None of these components alone is sufficient. The generator cannot check its own work. The aesthetic critic cannot detect tactical violations. The analytic critic cannot appreciate structural elegance. The verifier cannot detect impending violations. The attacker finds flaws but cannot build. The coordinator adds no new knowledge but is essential to integration. Their coordination is what produces depth. Just as multicellular organisms achieve what single cells cannot, this architecture achieves constraint satisfaction not through any single agent’s capability but through the coordination of specialists, each covering the blind spots of the others.

This division of labour mirrors Dual-Process Theory in cognitive psychology [16]. The Generator and Aesthetic Critic operate as “System 1”: fast, intuitive, and pattern-based, relying on chunked representations and holistic evaluation. The Analytic Critic and Deterministic Verifier operate as “System 2”: slow, deliberative, and rule-bound, performing sequential constraint verification. Current monolithic LLMs attempt to force System 1 architecture to perform System 2 tasks, resulting in the shallow reasoning diagnosed in Section 2.

Figure A.3 shows the full architecture of this Society of Specialised LLMs.

6.1. Coordinator Search Procedure (CBS-DL)

The following specification makes the Coordinator’s operation precise. The algorithm is presented as a prose specification to capture the full control logic; it can be directly implemented from this description.

Coordinator Search Procedure (Constrained Beam Search with Depth Limit, CBS-DL).

The coordinator maintains a beam of up to w partial solutions, each a sequence of reasoning chunks produced by the Generator. The search proceeds step-by-step, with each branch limited to a maximum trace length $D_{\max}$.

Initialisation. The beam is initialised with $\{s_0\}$.

Expansion. At each step, for every state s in the current beam, the Generator is called m times independently, yielding m candidate reasoning chunks. Each chunk is appended to s , forming an extended state s' . *Budget:* Each call to Generate consumes one unit of the compute budget B .

Hard pruning. Each candidate s' is evaluated by the Deterministic Verifier. The verifier checks all hard constraints on the current partial state; it does not require completeness. If any constraint is violated, the candidate is discarded. *Budget:* Each verifier invocation consumes one unit.

Adversarial probing and repair. For each surviving candidate, the Adversarial Attacker probes it, returning a set of counterexamples C . *Budget:* Each probing round consumes one unit.

If C is empty, the candidate proceeds directly to scoring. If C is non-empty, the system attempts repair using a snapshot-driven loop:

1. Create a snapshot $S \leftarrow C$. (S is the set of counterexamples to address.)
2. $\text{repair_attempts} \leftarrow 0$.
3. **while** $S \neq \emptyset$ **and** $\text{repair_attempts} < R$:
 - (a) Pick a counterexample c from S .
 - (b) $s_{\text{rep}} \leftarrow \text{RollbackAndRepair}(s', c)$.
 - (c) $\text{repair_attempts} \leftarrow \text{repair_attempts} + 1$.
 - (d) *Budget:* The repair call consumes one unit.
 - (e) **If** s_{rep} is NULL **or** re-verification of s_{rep} fails:
 - i. Remove c from S . (*This counterexample is unfixable.*)
 - ii. Continue the while loop.
 - (f) **Else** (*Repair produced a verifier-valid state*):
 - i. $s' \leftarrow s_{\text{rep}}$.

- ii. Re-probe s' with the Attacker, obtaining a fresh counterexample set C' .
 - iii. *Budget*: The re-probe consumes one unit.
 - iv. **If** $C' = \emptyset$: (All counterexamples resolved.) $C \leftarrow \emptyset$, break the while loop.
 - v. **Else**: (New counterexamples discovered; restart from a fresh snapshot.) $C \leftarrow C'$, $S \leftarrow C'$.
4. **End while.**
5. (Post-loop discard rule.) **If** $C \neq \emptyset$: Discard the candidate. (Counterexamples remain after all repair attempts are exhausted.) **Else**: The candidate is attacker-clean and proceeds to scoring.

Scoring. Each surviving, attacker-clean candidate is scored using the bidirectional objective:

$$\text{score}(s') = \alpha \cdot V_{\text{fwd}}(s') - \beta \cdot V_{\text{bwd}}(s'), \quad (2)$$

where V_{fwd} is the Aesthetic Critic’s success probability estimate (reward) and V_{bwd} the Analytic Critic’s constraint violation cost estimate (penalty). *Budget*: Both critic evaluations together consume one unit.

Beam selection. From the pool of all scored candidates across all states in the current beam, the top w by score are selected to form the beam for the next step. If fewer than w candidates survive, all are kept.

Termination. After beam selection, each state in the new beam is tested for completeness. A state is considered complete when all problem-specific completeness criteria are satisfied (e.g., every required constraint has been explicitly addressed and the solution is syntactically finalised). The search returns the **first** complete state that also passes both the Deterministic Verifier and the Adversarial Attacker.

If the beam becomes empty, the compute budget B is exhausted, or the maximum trace length D_{max} is reached without finding a complete solution, the search returns FAIL.

The early-exit policy (returning the first complete, fully verified solution) reflects the design choice that any constraint-satisfying complete solution is equally acceptable; the critics’ scores serve solely to steer the search, not to rank final answers.

Remark (Motivating principle for bidirectional search). If the Analytic Critic provides an admissible heuristic—one that never overestimates the remaining constraint pressure—then, in an unconstrained (exhaustive) search, classical A* results (Hart et al., 1968 [36]) would guarantee optimal node expansion. CBS-DL is a beam search with finite width w ; it does not inherit this guarantee. The heuristic serves as a gradient for node ordering, not a formal completeness or optimality property. The practical utility of the architecture rests on how well the learned critics approximate these ideal conditions.

6.2. Constraint State Management

The critics and attackers rely on a compact, globally consistent view of the constraints, which the autoregressive generator cannot maintain internally. To provide this view without requiring the critics to attend over the full trace, the Coordinator maintains a **Constraint State Register (CSR)**. The CSR is a structured record that encodes, for each constraint, (a) its current status (satisfied, violated, or unaddressed), and (b) the *partial design decisions* that are causally relevant to that constraint’s eventual satisfaction. For example, in the software-architecture task of Section 4, the CSR for a “strict dependency acyclicity” constraint would store the set of declared dependencies to date, not merely a flag; the CSR for a “formal API contract adherence” constraint would store the current function signatures and their types.

Because the CSR grows with the number of *entities* introduced during reasoning (modules, functions, variables), not with the length of the trace, its size is bounded by the problem’s structural complexity rather than by the generation horizon. For a task with m constraints and n domain entities, the CSR occupies $O(m + n)$ space, independent of the number of reasoning steps. The Coordinator updates the CSR after each reasoning step by querying the Deterministic Verifier on each constraint and extracting the updated

partial state; the Verifier is already performing the symbolic checks, so this extraction is a by-product of verification, not a separate cost.

The Analytic Critic receives the CSR together with the new candidate chunk s' . The CSR provides the critic with the causal context needed to assess whether the candidate chunk creates an emerging contradiction: for example, whether a newly proposed threading model conflicts with the memory-allocation strategy recorded in the CSR. The critic then computes its constraint-pressure score by comparing the CSR with the desired terminal state and by detecting local inconsistencies between the candidate chunk and the partial state stored in the CSR.

We now give a formal characterisation of the CSR’s informational sufficiency. The result shows that, under a natural design condition on the reasoning process, the compact register is provably sufficient for predicting all eventual constraint violations, and it is the coarsest statistic with this universal property.

Theorem 3 (CSR Characterisation for Outcome-Markov Processes). Consider a reasoning process over a finite set of variables with a finite domain, terminating with eventual state $V = (\sigma_\infty, R_\infty)$. Define the Constraint State Register $\text{CSR}_k = (\sigma_k, D_k, R_k, \{\text{status}_{j=1}^m\})$, where each status is a deterministic function of (σ_k, D_k, R_k) . The process is called **outcome-Markov** if the conditional distribution of the eventual state given the full trace history depends only on the current partial state: $\Pr(V \in \cdot \mid T_k) = \Pr(V \in \cdot \mid \sigma_k, D_k, R_k)$.

(A) Sufficiency. If the process is outcome-Markov, then for every constraint c_j with violation indicator E_j , and every step k :

$$\Pr(E_j \mid T_k) = \Pr(E_j \mid \text{CSR}_k). \quad (3)$$

More generally, for any measurable function ϕ of the eventual state,

$$\mathbb{E}[\phi(V) \mid T_k] = \mathbb{E}[\phi(V) \mid \text{CSR}_k]. \quad (4)$$

(B) Minimality. Let S be any statistic of the trace. If S is universally sufficient for prediction—i.e., for every outcome-Markov process and every trace, $\Pr(V \in \cdot \mid T_k) = \Pr(V \in \cdot \mid S(T_k))$ —then S must determine the CSR: for any two traces T_k, T'_k , $S(T_k) = S(T'_k)$ implies $\text{CSR}_k(T_k) = \text{CSR}_k(T'_k)$. The CSR is therefore the coarsest universally sufficient statistic up to deterministic equivalence.

Proof sketch. **(A)** By construction, the CSR contains (σ_k, D_k, R_k) and the statuses, each of which is a deterministic function of that triple. Hence conditioning on the CSR is equivalent to conditioning on (σ_k, D_k, R_k) . The outcome-Markov property directly gives $\Pr(V \in \cdot \mid T_k) = \Pr(V \in \cdot \mid \sigma_k, D_k, R_k)$. Constraint violations are deterministic functions of V ; the generalisation to arbitrary measurable ϕ follows by the tower property of conditional expectation.

(B) Assume S does not determine the CSR. Then there exist traces T_k, T'_k with $S(T_k) = S(T'_k)$ but distinct CSR values $c \neq c'$. Construct an immediate-termination process: at step k the process halts and emits the current state as V , assigning positive probability to both traces. This process is outcome-Markov. By universal sufficiency, $\Pr(V = c \mid S = s^*) = 1$ (from T_k) and $\Pr(V = c' \mid S = s^*) = 1$ (from T'_k), a contradiction since $c \neq c'$. Therefore S must determine the CSR. $\square$

Scope and caveats. Theorem 3 is an information-theoretic specification, not an implementation guarantee. It assumes exact CSR extraction, outcome-Markovianity of the reasoning process, and concerns only properties of the eventual state. Path-dependent control properties—such as the number of backtracks performed, the remaining compute budget, or the depth of the current beam—are not functions of V alone and require additional algorithmic state in the Coordinator (Section 6.1). The universal minimality result is a worst-case lower bound: for a specific task and a specific Generator, a coarser register may suffice. The theorem identifies the common denominator across all tasks, providing a template that the architect instantiates per domain.

Causal attribution for repair. The RollbackAndRepair mechanism in the CBS-DL specification (Section 6.1) additionally requires the system to identify which prior decision caused a detected counterexample—a causal attribution problem. If the CSR omits the relevant decision from its compressed representation, repair may fail even when a fix exists. This places a further burden on the CSR design: it must encode not only the status of each constraint but the specific design choices that causally support that status, so that the repair module can backtrack to the appropriate point. Automated causal attribution in long reasoning traces is an open research problem, and the current architecture relies on manual CSR engineering to make the causal links explicit.

Similarly, the Aesthetic Critic receives a summary graph connectivity score derived from the conceptual graph built by the Coordinator. The conceptual graph is extracted from the reasoning trace using a combination of dependency parsing (for formal reasoning) and discourse parsing (for rhetorical structure); its nodes and edges are symbolic and can be updated incrementally. The algebraic connectivity and orphan-node count are computed symbolically, bypassing neural attention limits. For domains where formal parsing is unavailable, an LLM-based extraction step introduces the same attention dilution the architecture aims to avoid; the current design is therefore most effective for tasks with machine-parseable reasoning structure.

6.3. The Neuro-Symbolic Translation Bottleneck and its Fractal Resolution

A critical challenge in any neuro-symbolic architecture is the translation of natural-language reasoning into a formal specification that a deterministic verifier can check. If the translation step is performed by a fallible neural model, the verifier may certify a mistranslation rather than the original argument, undermining the constitutive guarantee.

The Society of Specialised LLMs addresses this bottleneck by applying its own principle recursively. The translation step is treated as a sub-problem—a local constraint-satisfaction task—and delegated to a dedicated **Translator**. The Translator is itself an instance of the same six-role architecture: a Generator proposes candidate formal encodings; Aesthetic and Analytic Critics evaluate their fidelity and structural coherence; a Deterministic Verifier checks syntactic correctness and completeness; and a Coordinator selects the best encoding. If the domain allows, the Translator’s output can be further verified by a lower-level instance of the society, recursing down to a level where the translation is trivially formal (e.g., a simple parser). This fractal recursion ensures that the translation step is subject to the same verification guarantees—Theorems 2* and 3—at every scale.

The Translator is not a new component type; it is a re-instantiation of the existing six roles, specialised to the domain of formal encoding. The architecture thus does not rely on a single fallible neural translation step; it recursively verifies translations to whatever depth is required for the domain. The constitutive guarantee is preserved: any formal specification that emerges from this cascade has been verified at every level. The Verifier guarantees only the syntactic correctness of the translation, not its semantic fidelity to the original natural-language intent. Semantic alignment is evaluated heuristically by the Critics and Attacker within the translation society, and ultimately validated by the external grounding provided by the human operator or environmental feedback.

6.4. Efficiency of Hard Pruning (Illustrative Model)

The following simplified model is provided solely for pedagogical intuition; its independence assumption directly contradicts the interlocking-constraint setting, and the baseline is a strawman. It demonstrates the value of early pruning, not the full architecture’s efficiency.

Simplified model. Assume each generated reasoning chunk is independently valid with probability $p \in (0, 1)$. The baseline autoregressive model generates a sequence of L chunks; if any chunk is invalid, the entire sequence is discarded and regenerated from scratch. CBS-DL with beam width w and expansion

multiplicity $m = w$ generates at most w chunks per state per depth; invalid chunks are pruned immediately and never propagated.

Claim. Under the above assumptions, the expected total number of chunks generated by the baseline is $\mathbb{E}[N_{\text{base}}] = L/p^L$ (exponential in L), while for CBS-DL it is bounded by $\mathbb{E}[N_{\text{CBS}}] \leq Lw/p$ (linear in L).

Justification. For the baseline: a sequence is valid with probability p^L ; the number of trials is geometric with mean $1/p^L$; each trial generates L chunks. For CBS-DL: at each depth, at most w states exist; each state generates chunks until a valid one appears, with a geometric number of attempts per state of mean $1/p$; summing over w states and L depths gives the bound.

Limitations. This model assumes independent validity across chunks, which is false for interlocking constraints. It also omits the repair loop, which can recover some invalid chunks without resampling. The value of the model is purely pedagogical: it illustrates the exponential-to-linear contrast produced by hard pruning alone. A rigorous treatment with realistic error dependencies is left for future work.

6.5. Probabilistic Advantage of Constrained Beam Search

The illustrative model in Section 6.4 assumed independent chunk validity, which is unrealistic for interlocking constraints. We now prove a stronger result that explicitly models the interaction between Generator and Critics. The result shows that the Coordinator’s beam search amplifies the probability of finding a valid complete solution, even when the search space contains dead ends, provided the Critics can distinguish extendable continuations from dead ends with better-than-chance accuracy.

Setup. A reasoning task requires a sequence of L chunks, all of which must be valid. A partial solution of length k is *valid* if all k chunks satisfy the hard constraints up to that point, and *extendable* if it can be completed to a valid full solution (probability > 0). The Coordinator performs CBS-DL with beam width $w \geq 1$ and expansion multiplicity $m = w$. The Deterministic Verifier is perfect: only valid candidates survive hard pruning.

Assumptions. We make two assumptions, both of which are explicit design targets for the architecture.

1. **Generator quality.** When the Generator is prompted with any extendable prefix, each of the w independently generated candidate chunks is **both valid and extendable** with probability at least $q > 0$. The parameter q captures the Generator’s ability—possibly after fine-tuning—to produce locally correct chunks that do not create semantic dead ends.
2. **Critic discriminability.** After hard pruning, the Coordinator scores every valid candidate. Let C be the set of all scored valid candidates generated from all prefixes in the beam (at most w^2 candidates), and let X be the number of candidates in C that are extendable and valid. The Critics satisfy the following retention property: if $X \geq 1$, then with probability at least $\theta > 0.5$ the highest-scored extendable valid candidate is among the top- w candidates selected for the next beam. The parameter θ captures the Critics’ joint ability—via the bidirectional scoring function—to rank globally survivable continuations above dead ends.

Baseline (Monolithic generation). The baseline model generates a full sequence of L chunks in one autoregressive pass, without external verification or backtracking. From the weak-link constraint, even a single chunk’s validity is capped by the weakest cognitive requirement. In the worst case—where all requirements except fluency are severely neglected—the monolithic success probability can be arbitrarily close to zero for any non-trivial L .

Theorem 2* (Success Probability Amplification with Critic-Guided Retention). Under the above assumptions, the probability that the Coordinator produces at least one complete valid solution of length L satisfies

$$P_{\text{CBS}} \geq (\theta \cdot (1 - (1 - q)^w))^L. \quad (5)$$

If q and θ are constant across depths, the right-hand side provides a single exponential decay rate. The architecture amplifies the per-step success probability whenever $\theta \cdot (1 - (1 - q)^w) > q$. This condition holds when the Generator’s standalone success probability q is small enough that the diversity gain from beam search outweighs the noise introduced by an imperfect critic. When the critic is nearly perfect ($\theta \approx 1$), amplification is guaranteed for any $q \in (0, 1)$ and any $w \geq 2$; when the critic is weak, the architecture may perform worse than the monolithic baseline, and the condition identifies the precise threshold above which diversification is beneficial.

Proof sketch. Induction on depth. Let E_k be the event that the beam at depth k contains at least one extendable prefix. E_0 holds trivially. Condition on E_{k-1} . Consider the worst-case beam configuration: exactly one extendable prefix, the rest dead ends. Expanding that prefix gives w independent candidates, each extendable-and-valid with probability $\geq q$. Hence $\Pr(X \geq 1) \geq 1 - (1 - q)^w$. Conditioned on $X \geq 1$, the retention property guarantees survival of an extendable candidate with probability $\geq \theta$. Therefore $\Pr(E_k | E_{k-1}) \geq \theta \cdot (1 - (1 - q)^w)$ in the worst-case configuration, and the bound holds for all configurations. The induction yields $P(E_L) \geq (\theta \cdot (1 - (1 - q)^w))^L$. Since an extendable prefix of length L is a complete valid solution, the theorem follows. A full formal proof is provided in Appendix A. $\square$

Discussion. Theorem 2* captures the division of labour between Generator and Critics. The Generator must produce continuations that are locally valid and globally survivable (q); the Critics must recognise them (θ). Neither component alone suffices, but together they amplify the per-step success probability from q to approximately $\theta \cdot (1 - (1 - q)^w)$. This amplification is most significant when individual chunks are hard to get right (small q), precisely the setting where interlocking constraints are most challenging.

Unlike the original Theorem 2 of Abaee (2026), this version does not assume the absence of dead ends. Dead ends may exist; the Critics’ role is to avoid retaining them when extendable alternatives are generated. The bound is parameterised by critic quality θ : when critics are perfect ($\theta = 1$) and the Generator’s extendable-valid probability q is high, the bound approaches $(1 - (1 - q)^w)^L$, matching the idealised amplification of the original analysis. When critics are imperfect, the bound degrades gracefully.

The theorem also clarifies the relationship between the search architecture and compute budget. For a fixed budget B scaling with beam width w , the compute-efficiency ratio $R(B) = P_{\text{CBS}}(B)/P_{\text{mono}}(B)$ grows exponentially in L whenever q is not the unique weakest link in the monolithic system. This reframes the search-depth concern of Section 7: the architecture does not require chess-engine depths to provide gains; it provides the best possible return on whatever compute budget is available.

The consciousness link: Global Workspace Theory. This proposed architecture structurally parallels Global Workspace Theory (GWT), a leading framework in cognitive science for understanding how specialised brain regions integrate information [17, 18]. In GWT, unconscious, specialised processors compete for access to a central workspace; the winning coalition is broadcast globally, enabling flexible, deliberate action. The Society of Specialised LLMs implements a structurally analogous mechanism: the Generator, Critics, and Attacker compete via the Coordinator’s scoring function for the right to propose the next reasoning step; the selected candidate is broadcast through the shared CSR and becomes the focus of the next cycle. This is a functional parallel to the workspace architecture, not a claim about phenomenal consciousness. The architecture provides the mechanistic prerequisites for a system to monitor its own trajectory and intervene on contradictions—a form of access consciousness in the sense of Block (1995) [19].

This architecture is not speculative in its individual parts. Process-based reward models [20, 21] are early versions of the analytic critic. Tree-of-thought prompting [22] and Monte Carlo tree search for language models [23] are early versions of the coordinator. Constraint-satisfaction benchmarks [24] confirm that advanced models struggle with long-chain reasoning, underscoring the need for verifier-anchored architectures. Self-play and debate [25, 26] with adversarial attackers have shown promise [27, 28]. The

aesthetic critic is the least explored component, but its function—holistic pattern evaluation anchored to outcome correlation—is directly analogous to the neural evaluation functions that transformed computer chess. The deterministic verifier is standard software. What is novel is the proposal to integrate them into a single, coordinated system organised by the principle that specialised intelligences covering each other’s blind spots outperform any monolithic model.

7. Limitations of the analogy

The chess analogy illuminates an incentive and architectural problem. It does not, on its own, solve the hard limits that remain.

Search depth as a compute-optimality trade-off. A chess engine searches millions of positions, and current LLMs cannot match that scale. This does not contradict the architecture’s value; it characterises a continuum. Theorem 2* shows that even modest beam widths and depths yield exponential gains over monolithic generation, and the architecture scales gracefully with available compute. The compute-efficiency ratio $R(B)$ introduced in Section 6.5 grows with the beam width w , which is itself bounded by the budget B . The architecture thus reframes search depth from a binary “can or cannot” to a principled trade-off: any compute budget can be allocated optimally to maximise the probability of constraint satisfaction, and the gains from additional compute are subject to diminishing returns, exactly as in chess engines.

Finite context and attention dilution. Current models have bounded context windows and attention that decays with distance [4]. For sufficiently complex tasks, the sheer number of constraints may exceed the model’s ability to hold them in simultaneous attention. A constraint can be functionally invisible—not because the model devalues it, but because the architecture cannot access it.

The epistemic void for genuinely open-ended tasks. For tasks where constraints cannot be formalised—*aesthetic judgment, open-ended philosophical speculation, creative design*—the analogy breaks down. There is no “rule set” to internalise, no “legal move” to check. Any “king-safety” signal for such tasks would itself be a human-designed proxy, vulnerable to the same rigidity and blind spots that plagued early handcrafted chess heuristics, and to Goodhart’s Law. The proposed research programme is designed for tasks with specifiable, checkable constraints. It does not claim to solve open-ended reasoning where the standard of correctness is itself contested. For rich interlocking constraints, the CSR must be manually engineered per domain; fully general reasoning with arbitrary semantic constraints is beyond the current scope.

The illusion of self-awareness. While the proposed architecture implements a workspace-style integration mechanism structurally parallel to GWT, this is distinct from genuine phenomenal consciousness [19]. Furthermore, a system with metacognitive critics can still suffer from “introspective hallucination” or multi-agent collusion. While multi-agent debate has shown significant promise for improving factuality [29], it is a well-documented risk in the literature that if agents share the same foundational biases, they can confidently converge on incorrect answers, creating an echo chamber of mutually reinforcing delusions. Without the deterministic verifier to anchor the system to reality, the society of LLMs risks becoming just such an echo chamber—highly articulate, but detached from truth. Architecture provides the machinery for self-reflection, but it does not guarantee truth.

Thinking time is not depth. Even with topological metrics, a model can produce backtracking-like patterns without genuine self-correction. Chain-of-thought explanations can be unfaithful to the model’s actual reasoning process [30, 31, 32]. A rigorous, causal metric for reasoning depth remains to be developed.

These limits are severe. But they are also precise. The chess analogy tells us exactly where the proposed solutions apply—to constraint-bound reasoning tasks—and where they do not.

8. The move we need to make

Given these boundaries, what should we actually do? The pawn-grabbing problem is a direct consequence of the incentives we have built and the architectures we have chosen. If we continue to optimise for local fluency and short-term helpfulness, with single-path generation that cannot check its own work, we should expect models that are increasingly eloquent and decreasingly trustworthy.

The chess analogy suggests a different path: treat constraint satisfaction as constitutive of generation—a hard architectural requirement, not a soft reward signal. This requires building systems with differentiated components—a generator, an aesthetic critic for structural harmony, an analytic critic for constraint pressure, a deterministic verifier, an adversarial attacker, and a coordinator, with the neuro-symbolic translation step handled by a recursive instantiation of the same six-role architecture—each with its own mode of knowing, each blind to what the others see, their coordination producing what no monolithic model can achieve. This is not a call for larger models. It is a call for architectures that reflect the principle, demonstrated by chess engines and paralleled in biology, that complex intelligence is not a single thing doing a single thing well. It is a network of specialists, covering each other’s blind spots, hunting together.

This architectural shift directly addresses the emerging paradigm of **inference-time compute scaling**. Recent research demonstrates that dynamically allocating more computational resources during inference—via extended search and verification—can yield greater reasoning improvements than simply scaling model parameters [33]. However, blindly generating more tokens is inefficient. The Society of Specialised LLMs provides the necessary scaffolding for *optimal* compute allocation. By using the Analytic Critic to identify high-pressure states (impending constraint violations), the Coordinator can dynamically allocate deeper search and more compute precisely where the reasoning trajectory is most fragile, rather than wasting inference compute on structurally sound, low-pressure text generation.

This architectural shift also suggests a path toward more general reasoning capabilities. Currently, the field often conflates a broad knowledge base with general intelligence. But a model that has memorised the internet yet reasons via myopic, single-path token prediction possesses narrow *reasoning mechanics*, even if its knowledge is general. True generality requires not just the capacity to retrieve any fact, but the structural capacity to sustain long-horizon, multi-constraint goal pursuit across domains where constraints can be specified. This architectural shift aligns with theoretical critiques of purely autoregressive intelligence. As argued by LeCun [34], true autonomous intelligence requires an objective-driven architecture capable of maintaining a world model, predicting the consequences of actions, and optimizing a cost function over long horizons—capabilities that next-token prediction fundamentally lacks. The society of specialised LLMs proposed here is a blueprint for generalising the *process* of deliberation. By decoupling the generation of ideas from the evaluation of their structural harmony and constraint satisfaction, we build a reasoning engine that can, in principle, be applied to any domain where constraints can be specified. Extending this to fully open-ended tasks remains a challenge for future work.

Future work. Several components of the proposed architecture are ripe for immediate development. The Analytic Critic can be trained using temporal-difference learning ($TD(\lambda)$) or step-level direct preference optimisation (Step-DPO), which would allow it to bootstrap distant constraint-violation signals back into dense, intermediate feedback—directly countering the Horizon Effect identified in Section 2. The Aesthetic Critic can be trained via graph contrastive learning (GraphCL) on conceptual connectivity graphs extracted from reasoning traces, maximising structural harmony without relying on subjective human annotations. For tasks where constraints can be formalised into decidable logical fragments, the Deterministic Verifier can be instantiated as an SMT solver (e.g., Z3) using the DPLL(T) algorithm, providing sound and complete checking at the cost of a neuro-symbolic translation step whose reliability is addressed by the fractal Translator architecture (Section 6.3). The Adversarial Attacker can be strengthened by sampling diverse

probes via determinantal point processes (DPPs), with the important caveat that DPPs enforce surface-level diversity, not the epistemic independence required for Condorcet-style guarantees—a gap that remains to be closed. The Coordinator’s compute-allocation policy can be made adaptive, widening or deepening search in regions of high constraint pressure, a direct extension of recent work on inference-time compute scaling [33]. Finally, automatic learning of sufficient CSR state representations for rich interlocking constraints is a critical open problem; manually engineered CSRs serve as a stopgap, but scaling to arbitrary domains will require representation learning techniques that discover causally relevant state summaries from data. Formalising the CSR as a Markov state abstraction [38] and proving that it preserves sufficiency for interlocking constraint domains is a concrete theoretical next step, building on recent results that characterise when learned abstractions retain the Markov property.

The chess engines did not become superhuman by scaling the pawn-grabbing heuristic. They diversified. They added evaluation terms, deepened search, and integrated specialised components, each addressing a bottleneck the others could not. The language models can do the same—but only if we stop investing in fluency alone and build systems whose cognitive functions are as diverse and coordinated as the pieces on a chessboard at the moment of a decisive attack.

Data and code availability

The full text of all meta-prompts described, the raw model outputs, and the thinking-time measurements are available from the author upon reasonable request. I invite replication, critique, and extension.

Competing interests

The author declares no competing interests.

Author contributions

Amin Abaee: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing.

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Ethics approval

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Appendix A. Full Proof of Theorem 2* (Success Probability Amplification with Critic-Guided Retention)

Theorem 2* (Success probability amplification with critic-guided retention). Let a reasoning task require a sequence of L chunks, all of which must be valid. A partial solution of length k is *valid* if all k chunks satisfy the hard constraints up to that point, and *extendable* if it can be completed to a valid full solution (probability > 0). Assume:

1. **(Generator quality).** When prompted with any extendable prefix, each of the w independently generated candidate chunks is both valid and extendable with probability at least $q > 0$.
2. **(Critic discriminability).** After hard pruning, let C be the set of all scored valid candidates (at most w^2), and let X be the number of extendable valid candidates in C . If $X \geq 1$, then with probability at least $\theta > 0.5$ the highest-scored extendable valid candidate is among the top- w candidates selected for the next beam.
3. **(Perfect Verifier).** The Deterministic Verifier accepts a candidate iff it is valid; no valid candidate is ever rejected, and no invalid candidate is ever accepted.

Then the probability that CBS-DL produces at least one complete valid solution of length L satisfies

$$P_{\text{CBS}} \geq (\theta \cdot (1 - (1 - q)^w))^L. \quad (\text{A.1})$$

Proof. We prove the lower bound (A.1) by induction on the depth k .

Notation and events

Let $\mathcal{B}_k$ denote the set of states (prefixes) in the beam after k reasoning steps. Define the event

$$E_k = \{\mathcal{B}_k \text{ contains at least one extendable prefix}\}. \quad (\text{A.1})$$

E_0 holds trivially (the empty prefix is extendable). E_L implies a complete valid solution, because an extendable prefix of length L is already a full sequence and must be valid.

We will prove by induction that $P(E_k) \geq c^k$ with $c = \theta \cdot (1 - (1 - q)^w)$.

Base case: $k = 0$

$P(E_0) = 1 = c^0$. The inequality holds.

Inductive step

Assume $P(E_{k-1}) \geq c^{k-1}$. By the law of total probability,

$$P(E_k) = P(E_k | E_{k-1})P(E_{k-1}) + P(E_k | \neg E_{k-1})P(\neg E_{k-1}) \geq P(E_k | E_{k-1})P(E_{k-1}). \quad (\text{A.2})$$

Condition on E_{k-1} , and consider the worst-case beam configuration consistent with E_{k-1} : exactly one extendable prefix in the beam, the remaining $w - 1$ slots occupied by dead ends (valid but unextendable). Expanding the single extendable prefix yields w independent Generator calls, each producing an extendable valid candidate with probability $\geq q$. The number X of extendable valid candidates among the valid candidates satisfies

$$\Pr(X \geq 1) \geq 1 - (1 - q)^w. \quad (\text{A.2})$$

Conditioned on $X \geq 1$, the retention property of the Critics guarantees that the highest-scored extendable valid candidate is among the top- w selected for the next beam with probability at least θ . Hence, for this worst-case configuration,

$$\Pr(E_k | E_{k-1}, \text{worst-case}) \geq \theta \cdot (1 - (1 - q)^w) = c. \quad (\text{A.3})$$

Because the bound c holds for the worst-case configuration, it holds for every configuration satisfying E_{k-1} . Therefore

$$\Pr(E_k | E_{k-1}) \geq c. \quad (\text{A.3})$$

Substituting (A.3) and the induction hypothesis into (A.2) yields

$$P(E_k) \geq c \cdot c^{k-1} = c^k. \quad (\text{A.4})$$

This completes the induction. Setting $k = L$ gives (A.1). $\square$

Remark on the baseline

For the monolithic baseline, the weak-link constraint gives that a single chunk's validity probability is at most $\min_i q_i$. In the worst case, $\min_i q_i$ can be arbitrarily small if any cognitive requirement is severely neglected. The diversified architecture avoids this min-bound by distributing requirements across the verification loop, and Theorem 2* quantifies the resulting amplification.

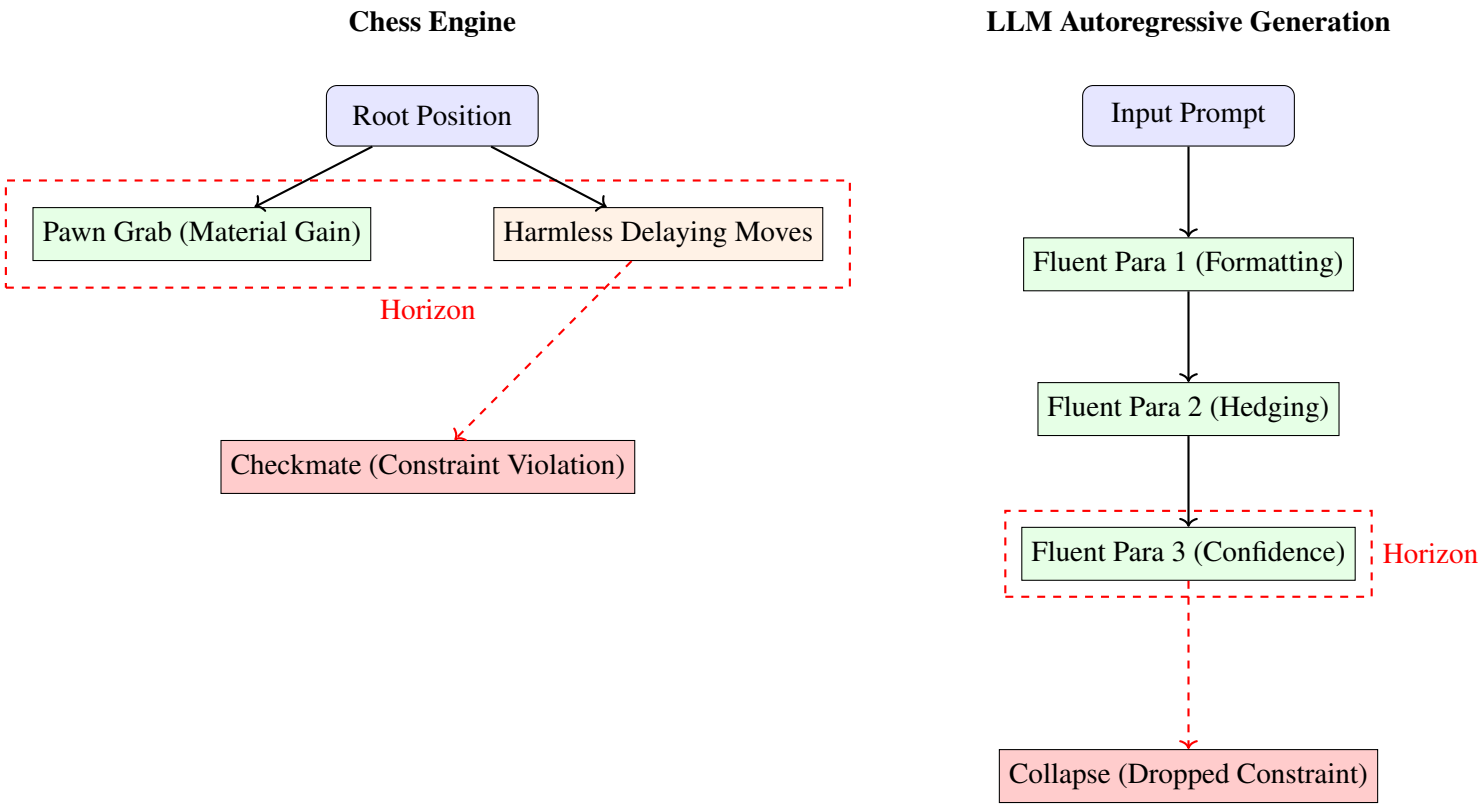
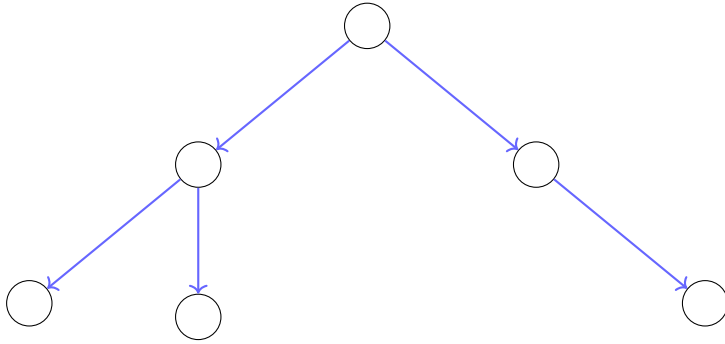


Figure A.1: The Linguistic Horizon Effect. Just as early chess engines pushed inevitable losses past their fixed search depth by grabbing immediate material, monolithic LLMs push constraint violations past their attention horizon by optimizing for local fluency and RLHF “pawns.”

A: Forward-Only Search



Forward (Fluency/Material)

B: Bidirectional Evaluation

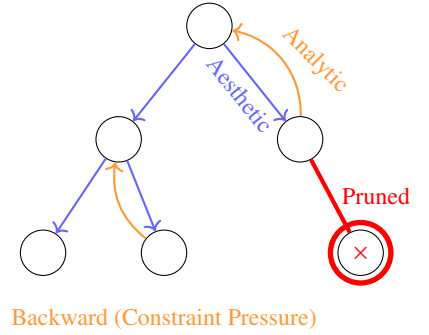


Figure A.2: Bidirectional evaluation in the Coordinator's search. The forward pass scores partial reasoning traces by structural harmony (piece mobility), while the backward pass scores them by accumulated constraint pressure (king safety). The search corridor is the region where both signals are favourable; the Coordinator navigates within this corridor.

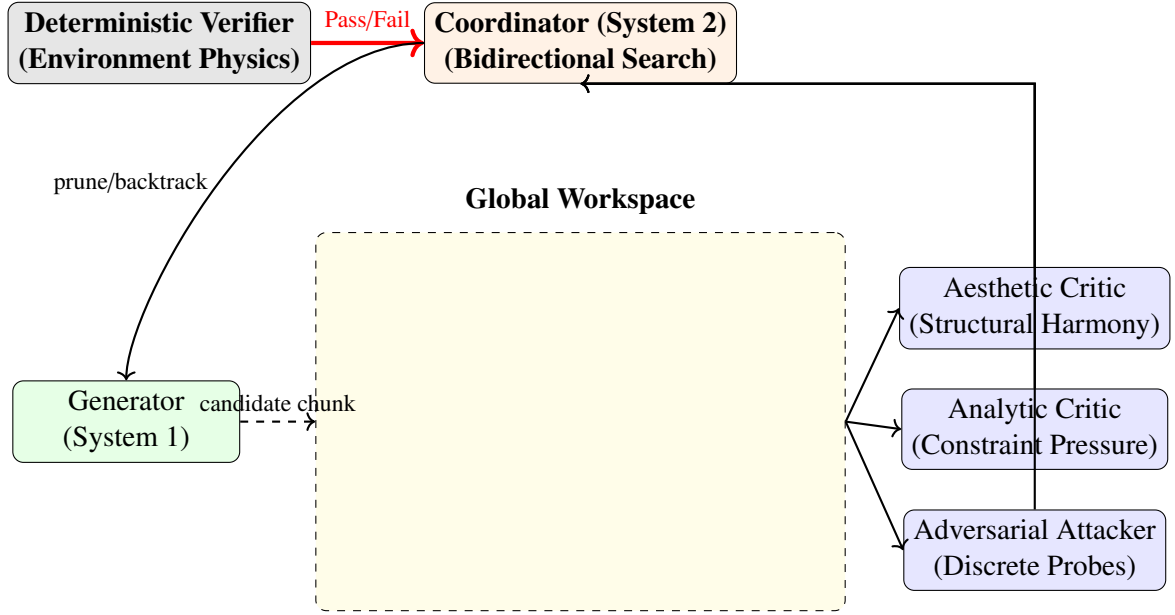


Figure A.3: The Society of Specialised LLMs architecture. Six roles—Generator, Aesthetic Critic, Analytic Critic, Deterministic Verifier, Adversarial Attacker, and Coordinator—interact through a shared workspace, with the neuro-symbolic Translator implemented as a recursive instantiation of the same six-role motif.

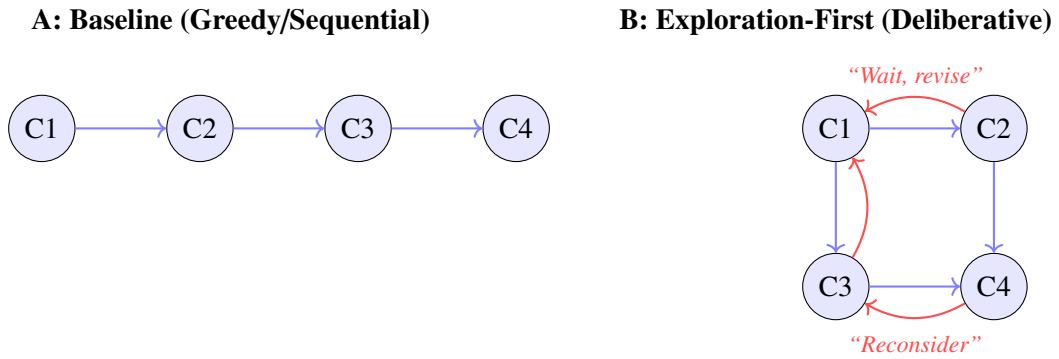


Figure A.4: Topological comparison of reasoning traces under baseline prompting (left) and the exploration-first protocol (right). Each node represents a defined concept or constraint; edges represent logical dependencies. The baseline trace exhibits isolated, sequential constraint handling with no backtracking; the exploration-first trace shows high connectivity, multiple constraint revisitations, and explicit backtracking events.